

WHEN EARTH MOVES



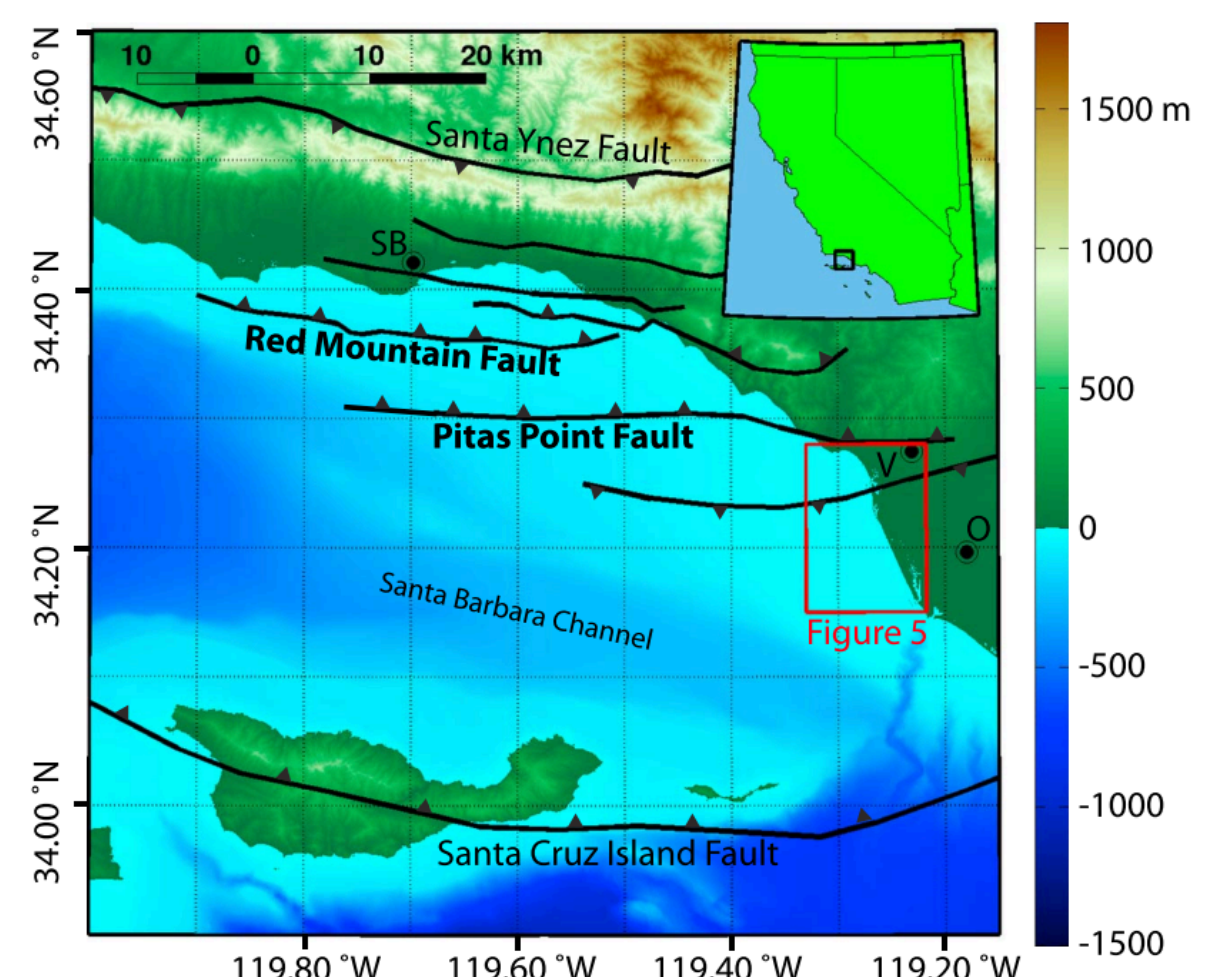
Abstract

Have you ever wondered how an earthquake under the ocean can create a giant wave? **Scientists studied a possible earthquake** near Ventura, California to understand how **tsunamis** might form and travel toward the coast. Using **computer models**, they simulated an underwater earthquake and were then able to

watch how the ocean responded. Their results showed that when the **seafloor suddenly moves** upward, it can push a huge amount of water with it and create many powerful tsunami waves. By studying possible tsunami **scenarios** scientists can help communities **prepare** and **stay safe**.

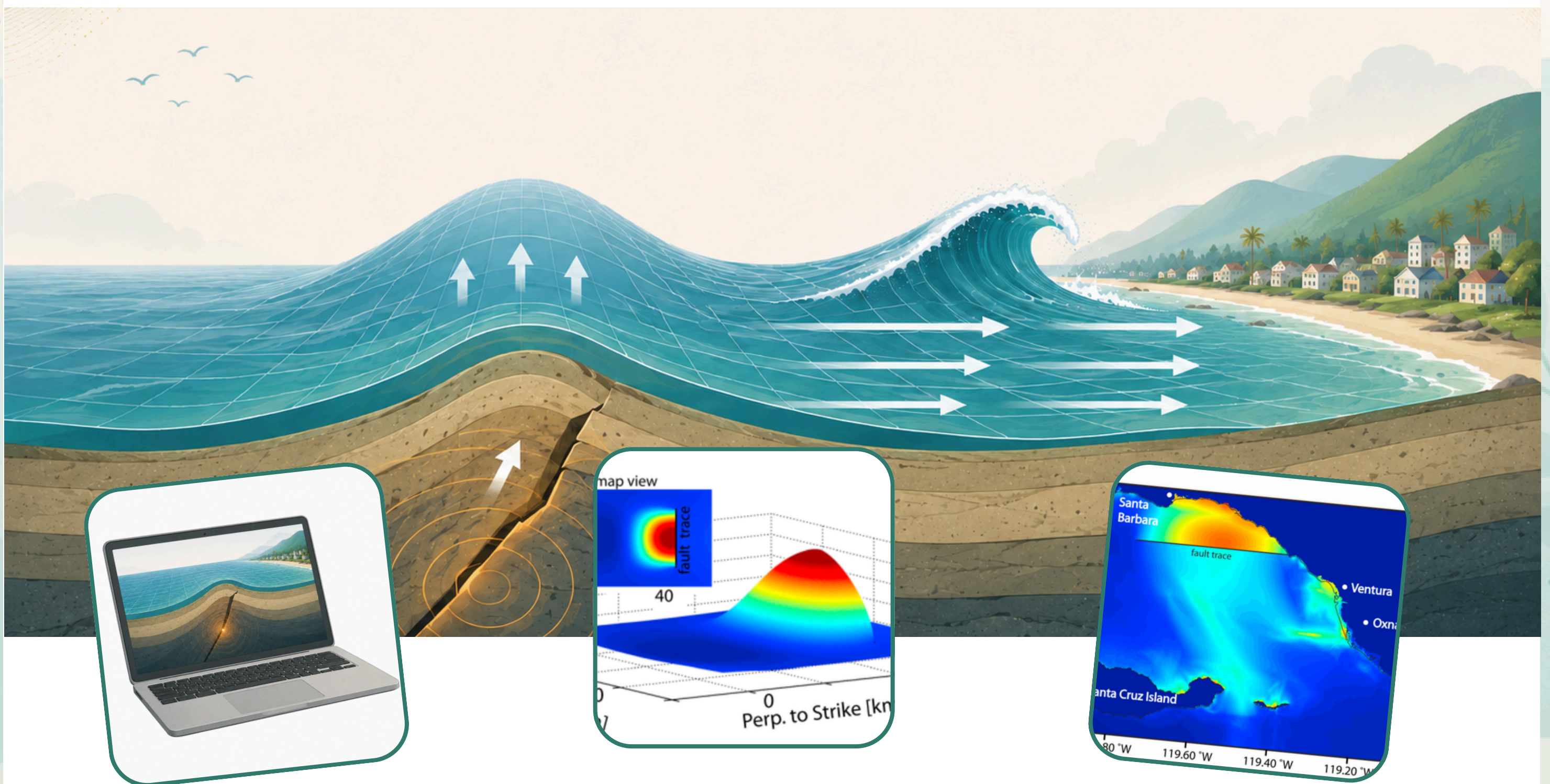
Introduction

The map shows **faults**, or cracks where Earth's crust can move, near **Southern California**. If a large underwater fault shifts the seafloor, it can **push ocean water** and create a **tsunami**. Scientists model these waves to help **coastal communities prepare**.



How do Scientists Model?

Scientists **cannot create real earthquakes** to test what might happen. Instead, they use **computer models**. In this study, **researchers modeled a possible large earthquake** beneath the **Santa Barbara Channel**, then used that model to see how the ocean might respond.



1. Build the earthquake model

Scientists created a computer model of a large earthquake on **offshore faults** near Ventura.

2. Calculate seafloor movement

The earthquake model showed how the ocean floor could **move up or down**.

3. Simulate the tsunami

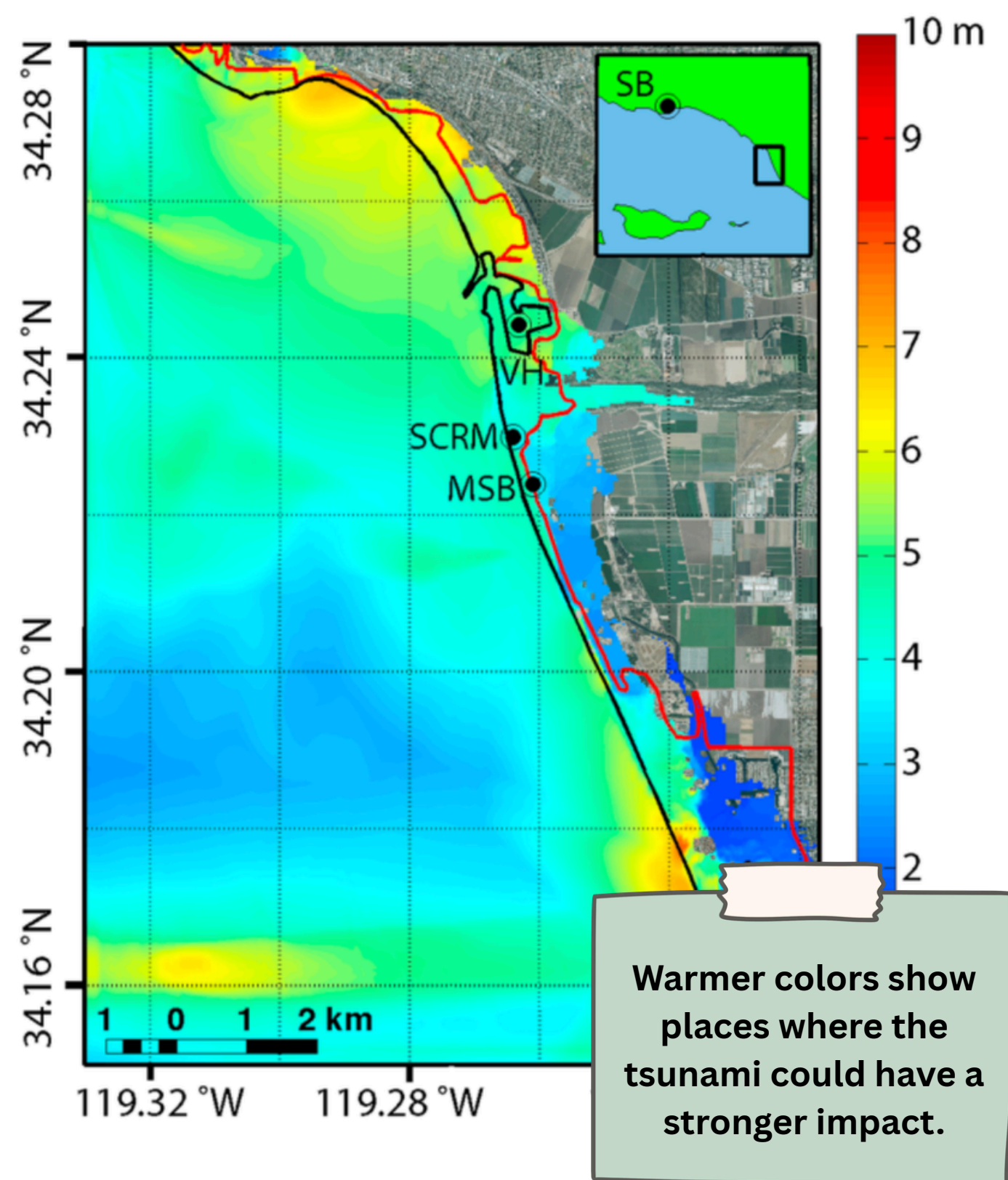
Scientists used the seafloor movement to model how **tsunami waves** might travel.



This was a scenario, not a prediction. Scientists were not saying exactly when an earthquake will happen. They were studying one possible event to better understand risk.

What Did They Discover?

The model showed that tsunami waves **would not affect every place equally**. Some waves moved **north** toward Santa Barbara, while others were bent and focused **east** toward Ventura and Oxnard. This happened because ocean **depth**, coastline **shape**, and wave **direction changed** how the tsunami energy moved.



Takeaways

Tsunami waves can change direction

Ocean depth and coastline shape can **bend** tsunami waves as they move.

Some places may be affected more.

The model showed **stronger** possible impacts near Santa Barbara, Ventura, and Oxnard.

Models help people prepare

Scientists use computer models to study possible disasters **before** they happen.

Key Terms

Fault — A crack in Earth’s crust where rocks can move.

Tsunami — A series of ocean waves caused by a sudden movement of water.

Computer model — A simulation scientists use to study possible events.

Scenario — One possible event scientists study. It is not a prediction.

Hazard — Something that could cause danger or damage.

Activity

Build a Mini Coastline

- Work in small groups to create a mini coastline using clay, paper, blocks, or small objects.
- Include features like a beach, town, harbor, or hill, and add water for the ocean.
- Create a wave from the “offshore fault” side of the tray and observe its impact on the coastline.
- Modify the coastline (e.g., add a harbor or barrier) and test again.

References

Ryan, K. J., Geist, E. L., Barall, M., & Oglesby, D. D. (2015). Dynamic models of an earthquake and tsunami offshore Ventura, California. *Geophysical Research Letters*, 42. <https://doi.org/10.1002/2015GL064507>